



Indefinite Integration : RACE-01

Subjective Type Question

Evaluate the following integrals :

1. (i) $\int 3x^{1/2} (1 + x^{3/2}) dx$ MII130
- (ii) $\int 2^x \cdot e^x dx$ MII131
- (iii) $\int \frac{e^{5/nx} - e^{4/nx}}{e^{3/nx} - e^{2/nx}} dx$ MII132
- (iv) $\int (e^{a/nx} + e^{x/na}) dx \ (a > 0)$ MII133
2. (i) $\int \frac{\sin^3 x \cos^3 x + \cos^4 x \sin^2 x}{\sin^2 x \cos^3 x} dx$ MII134
- (ii) $\int \frac{1 + \cos^2 x}{1 + \cos 2x} dx$ MII135
- (iii) $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$ MII136
- (iv) $I = \int \frac{dx}{\sin^2 x \cdot \cos^2 x}$ MII137
3. (i) $\int \frac{\sin^3 x + \cos^3 x}{1 - \sin x \cos x} dx$ MII138
- (ii) $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx$ MII139
- (iii) $\int \left(\sqrt{\frac{1+x}{1-x}} + \sqrt{\frac{1-x}{1+x}} \right) dx$ MII140
- (iv) $\int \frac{2 + 2x^2 - 3\sqrt{1-x^2}}{\sqrt{(1-x^2)(x^4 + 2x^2 + 1)}} dx$ MII141

4. (i) $\int \frac{2+3x^2}{x^2(1+x^2)} dx$ MII142
- (ii) $\int \frac{1+\tan^2 x}{1+\cot^2 x} dx$ MII143
- (iii) $\int \frac{x^6-1}{x^2+1} dx$ MII144
- (iv) $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$ MII145
5. (i) $\int \frac{dx}{1-\cos x}$ MII146
- (ii) $\int \frac{|x|\sqrt{x^2-1}+1+x^2}{|x|\sqrt{x^2-1}(1+x^2)} dx$ MII147
- (iii) $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$ MII148
- (iv) $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2\sin^2 2x} dx$ MII149
6. (i) $\int \frac{dx}{(2x-3)^5}$ MII150
- (ii) $\int \sqrt[5]{(8-3x)^6} dx$ MII151
- (iii) $\int e^{-3x+1} dx$ MII152
- (iv) $\int \frac{e^{2x}-1}{e^x} dx$ MII153
7. (i) $\int \frac{m}{\sqrt[3]{(a+bx)^2}} dx$ MII154
- (ii) $\int \frac{2x^3+3x^2+4x+5}{2x+1} dx$ MII155
- (iii) $\int \frac{2x+3}{3x+2} dx$ MII156
- (iv) $\int (e^x+1)^3 dx$ MII157

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8. (i) $\int \frac{x}{x^2 + 2x + 1} dx$ MII158
- (ii) $\int \cos x^\circ dx$ MII159
- (iii) $\int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$ MII160
- (iv) $\int \frac{1 - \cos x}{1 + \cos x} dx$ MII161
9. (i) $\int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^{-2} dx$ MII162
- (ii) $\int \frac{\cos x - \sin x}{\cos x + \sin x} (2 + 2 \sin 2x) dx$ MII163
- (iii) $\int \cos x \sin 3x dx$ MII164
- (iv) $\int (3 \sin x \cos^2 x - \sin^3 x) dx$ MII165
10. (i) $\int \sin x \cos x \cos 2x \cos 4x dx$ MII166
- (ii) $\int \left[\sin^2 \left(\frac{9\pi}{8} + \frac{x}{4} \right) - \sin^2 \left(\frac{7\pi}{8} + \frac{x}{4} \right) \right] dx$ MII167
- (iii) $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx$ MII168
11. (i) $\int \left[\frac{\cot^2 2x - 1}{2 \cot 2x} - \cos 8x \cot 4x \right] dx$ MII169
- (ii) $\int \cos x \cos 2x \cos 3x dx$ MII170
- (iii) $\int 4 \cos \frac{x}{2} \cdot \cos x \cdot \sin \frac{21}{2} x dx$ MII171
12. (i) $\int \frac{3}{(2x-1)(2x-3)} dx$ MII172
- (ii) $\int \frac{x+1}{(4x-1)(3x-2)} dx$ MII173
- (iii) $\int \frac{5x-2}{(2x+3)(3x-5)} dx$ MII174

ANSWER KEY

1. (i) $2x^{3/2} + x^3 + C$ (ii) $\frac{(2e)^x}{1+\ln 2} + C$ (iii) $\frac{x^3}{3} + C$ (iv) $\frac{x^{a+1}}{a+1} + \frac{a^x}{\ln a} + C$
2. (i) $-\cos x + \sin x + C$ (ii) $\frac{\tan x}{2} + \frac{x}{2} + C$ (iii) $-\cot x - \tan x + C$ (iv) $\tan x - \cot x + C$
3. (i) $-\cos x + \sin x + C$ (ii) $\sec x - \operatorname{cosec} x + C$ (iii) $2\sin^{-1} x + C$ (iv) $2\sin^{-1} x - 3\tan^{-1} x + C$
4. (i) $\frac{-2}{x} + \tan^{-1} x + C$ (ii) $\tan x - x + C$ (iii) $\frac{x^5}{5} - \frac{x^3}{3} + x - 2\tan^{-1} x + C$ (iv) $2\sin x + 2\cos \alpha \cdot x + C$
5. (i) $-\cot x - \operatorname{cosec} x + C$ (ii) $\tan^{-1} x + \sec^{-1} x + C$ (iii) $\tan x - \cot x - 3x + C$ (iv) $-2\cos x + C$
6. (i) $\frac{-1}{8(2x-3)^4} + C$ (ii) $\frac{-5}{33}(8-3x)^{11/5} + C$ (iii) $\frac{e^{-3x+1}}{-3} + C$ (iv) $e^x + e^{-x} + C$
7. (i) $\frac{3m}{b}(a+bx)^{1/3} + C$ (ii) $\frac{x^3}{3} + \frac{x^2}{2} + \frac{3x}{2} + \frac{7\ln(2x+1)}{4} + C$
(iii) $\frac{2x}{3} + \frac{5}{9}\ln(3x+2) + C$ (iv) $\frac{e^{3x}}{3} + \frac{3e^{2x}}{2} + 3e^x + x + C$
8. (i) $\ln(x+1) + \frac{1}{x+1} + C$ (ii) $\frac{180}{\pi}\sin x^\circ + C$ (iii) $\frac{\sin 2x}{2} + C$ (iv) $2\tan \frac{x}{2} - x + C$
9. (i) $\frac{\tan\left(2x - \frac{\pi}{4}\right)}{2} + C$ (ii) $\sin 2x + C$ (iii) $\frac{1}{2}\int\left(-\frac{\cos 4x}{4} - \frac{\cos 2x}{2}\right) + C$ (iv) $\frac{-\cos 3x}{3} + C$
10. (i) $\frac{1}{8}\left(-\frac{\cos 8x}{8}\right) + C$ (ii) $\frac{1}{\sqrt{2}} \cdot 2\left(-\cos \frac{x}{2}\right) + C$ (iii) $\frac{1}{2}\left(-\frac{\cos 4x}{4}\right) + C$
11. (i) $-\frac{\cos 8x}{8} + C$ (ii) $\frac{1}{4}\left(x + \frac{\sin 6x}{6} + \frac{\sin 4x}{4} + \frac{\sin 2x}{2}\right) + C$
(iii) $\frac{-\cos 12x}{12} - \frac{\cos 9x}{9} - \frac{\cos 11x}{11} - \frac{\cos 10x}{10} + C$
12. (i) $\frac{3}{4}\ln \frac{2x-3}{2x-1} + C$ (ii) $\frac{\ln|3x-2|}{3} - \frac{\ln|4x-1|}{4} + C$ (iii) $\frac{\ln|3x-5|}{3} + \frac{\ln|2x+3|}{2} + C$



Indefinite Integration : RACE-02

Subjective Type Question

Evaluate the following integrals :

1. (i) $\int \frac{e^x dx}{e^x + 1}$ MII175
- (ii) $\int \frac{d(1+x^2)}{\sqrt{1+x^2}}$ MII176
- (iii) $\int \frac{d\left(\frac{x}{3}\right)}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$ MII177
- (iv) $\int \frac{d(1+\ln x)}{\cos^2(1+\ln x)}$ MII178
2. (i) $\int \frac{d(\arcsin x)}{\arcsin x}$ MII179
- (ii) $\int \frac{dx}{x \ln x}$ MII180
- (iii) $\int \frac{\sqrt{\ln x}}{x} dx$ MII181
- (iv) $\int \frac{e^x + 1}{e^x + x} dx$ MII182
3. (i) $\int \frac{e^x + \cos x}{e^x + \sin x} dx$ MII183
- (ii) $\int 2x\sqrt{x^2+1} dx$ MII184
- (iii) $\int \frac{(2x-3)dx}{x^2-3x+8}$ MII185
- (iv) $\int e^{-x^3} x^2 dx$ MII186

4. (i) $\int \frac{(6x-5)dx}{2\sqrt{3x^2-5x+6}}$ MII187
- (ii) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ MII188
- (iii) $\int \frac{x^2 dx}{x^3+1}$ MII189
- (iv) $\int \frac{x-(x+1)\ln(x+1)}{x(x+1)\ln(x+1)} dx$ MII190
5. (i) $\int \frac{e^{2x} dx}{e^{2x}+a^2}$ MII191
- (ii) $\int (x+1)\sqrt{x^2+2x} dx$ MII192
- (iii) $\int \frac{\sqrt{x} dx}{1+x^{\frac{3}{2}}}$ MII193
- (iv) $\int \frac{2x^5-3x^2}{1+3x^3-x^6} dx$ MII194
6. (i) $\int e^{2x^2+\ln x} dx$ MII195
- (ii) $\int x^2 \sqrt[5]{x^3+2} dx$ MII196
- (iii) $\int \frac{x^4 dx}{\sqrt{4+x^5}}$ MII197
- (iv) $\int \frac{(\arctan x)^n}{1+x^2} dx$ MII198
7. (i) $\int \frac{dx}{x \ln x \ln(\ln x)}$ MII199
- (ii) $\int \frac{x}{\sqrt{x+2}} dx$ MII200
- (iii) $\int \frac{x}{2x+1} dx$ MII201
- (iv) $\int \frac{e^x-1}{e^x+1} dx$ MII202

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| 8. | (i) | $\int \frac{x^4}{1-x} dx$ | MII203 |
| | (ii) | $\int \frac{x^3 dx}{\sqrt{1+x^2}}$ | MII204 |
| | (iii) | $\int x^x \ln(ex) dx$ | MII205 |
| | (iv) | $\int \frac{dx}{1+\sqrt{x+1}}$ | MII206 |
| 9. | (i) | $\int \frac{\sin 2x}{1+\cos^2 x} dx$ | MII207 |
| | (ii) | $\int \frac{x^5+3x^4-x^3+8x^2-x+8}{x^2+1} dx$ | MII208 |
| | (iii) | $\int \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} dx$ | MII209 |
| | (iv) | $\int \frac{dx}{1+\sqrt[3]{x+1}}$ | MII210 |
| 10. | (i) | $\int \left(\frac{x-1}{x^2+1} \right)^2 dx$ | MII211 |
| | (ii) | $\int \frac{dx}{(x+\sqrt{x^2-1})^2}$ | MII212 |
| | (iii) | $\int \frac{(\sqrt{x}+1)dx}{\sqrt{x}(\sqrt[3]{x}+1)}$ | MII213 |
| | (iv) | $\int \frac{x^2 dx}{(a+bx)^2}$ | MII214 |

ANSWER KEY

1. (i) $\ln(e^x + 1) + C$ (ii) $2\sqrt{1+x^2} + C$ (iii) $\arcsin \frac{x}{3} + C$ (iv) $\tan(1 + \ln x) + C$
2. (i) $\ln|\arcsin x| + C$ (ii) $\ln|\ln x| + C$ (iii) $\frac{2}{3}\sqrt{(\ln x)^3} + C$ (iv) $\ln|e^x + x| + C$
3. (i) $\ln(e^x + \sin x) + c$ (ii) $\frac{2}{3}\sqrt{(x^2+1)^3} + C$ (iii) $\ln(x^2 - 3x + 8) + C$ (iv) $C - \frac{1}{3}e^{-x^3}$
4. (i) $\sqrt{3x^2 - 5x + 6} + c$ (ii) $2e^{\sqrt{x}} + c$ (iii) $\frac{1}{3}\ln(x^3 + 1) + c$ (iv) $\ln\left(\frac{\ln(1+x)}{x}\right) + c$
5. (i) $\frac{1}{2}\ln(e^{2x} + a^2) + C$ (ii) $\frac{1}{3}\sqrt{(x^2 + 2x)^3} + c$ (iii) $\frac{2}{3}\ln(1 + x^{\frac{3}{2}}) + c$ (iv) $-\frac{1}{3}(1 + 3x^3 - x^6) + C$
6. (i) $\frac{1}{4}e^{2x^2} + c$ (ii) $\frac{5}{18}\sqrt[5]{(x^3 + 2)^6} + C$ (iii) $\frac{2}{5}\sqrt{4 + x^5} + C$ (iv) $\frac{(\arctan x)^3}{3} + C$
7. (i) $\int \frac{dt}{t} = \ln t = \ln(\ln(\ln x)) + c$ (ii) $\frac{2}{3}(x + 2)^{3/2} - 4(x + 2)^{1/2} + C$
 (iii) $\frac{1}{2}\left[x - \frac{\ln(2x+1)}{2}\right] + C$ (iv) $2\ln(e^{x/2} + e^{-x/2}) + C$
8. (i) $C - \frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{1}{2}x^2 - x - \ln|1 - x|$ (ii) $\frac{1}{3}(1 + x^2)^{\frac{3}{2}} - (1 + x^2)^{\frac{1}{2}} + C$
 (iii) $x^x + C$ (iv) $2\left[\sqrt{x+1} - \ln(1 + \sqrt{x+1})\right] + C$
9. (i) $-\ln(1 + \cos^2 x) + c$ (ii) $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2}\ln(x^2 + 1) + 3\tan^{-1} x + C$
 (iii) $\frac{x^2}{2} - x + C$ (iv) $\frac{3}{2}(x+1)^{3/2} - 3(x+1)^{1/3} + 3\ln|1 + \sqrt[3]{x+1}| + C$
10. (i) $\tan^{-1} x - \frac{1}{x^2 + 1} + c$ (ii) $\frac{2}{3}[x^3 - \sqrt{(x^2 - 1)^3}] - x + C$
 (iii) $6\left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2}\ln(1 + t^2) - \tan^{-1} t\right] + C$ where $t = x^{1/6}$
 (iv) $\frac{1}{b^2}x - \frac{a}{b^3}\ln(bx + a^2) + \frac{a^2}{b^2}\left(-\frac{1}{bx + a}\right)\frac{1}{b} + c$



Indefinite Integration : RACE-03

Subjective Type Question

Evaluate the following integrals :

1. (i) $\int \frac{\sin x dx}{(1 - \cos x)^2}$ MII215
- (ii) $\int \frac{\cos x dx}{\sqrt[3]{\sin^2 x}}$ MII216
- (iii) $\int \frac{dx}{\cos^2 x \sqrt{1 + \tan x}}$ MII217
- (iv) $\int (\tan^3 x - x \tan^2 x) dx$ MII218
2. (i) $\int \frac{e^{\cot x} + e^{-\cot x}}{1 - \cos^2 x} dx$ MII219
- (ii) $\int \frac{1 - \sin x}{\cos x} dx$ MII220
- (iii) $\int \tan^3 x dx$ MII221
- (iv) $\int \frac{\sin 2x}{(a \sin^2 x + b \cos^2 x)^2} dx$ MII222
3. (i) $\int \frac{\sin^3 x}{\cos x} dx$ MII223
- (ii) $\int \frac{\cos^3 x dx}{\sin^4 x}$ MII224
- (iii) $\int \frac{\sin^3 \alpha}{\sqrt{\cos \alpha}} d\alpha$ MII225
- (iv) $\int \frac{1 + \sin x}{1 - \sin x} dx$ MII226

4. (i) $\int \left(\frac{x \cos x - \sin x}{x \sin x} \right) dx$ MII227
- (ii) $\int \cot (2x + 1) dx$ MII228
- (iii) $\int \tan 3x dx$ MII229
- (iv) $\int \frac{\sin^2 x + \cos x}{\sin x \cos x} dx$ MII230
5. (i) $\int \frac{\sin x + \cos x}{\sin x \cos x} dx$ MII231
- (ii) $\int \frac{\sin 5x}{\cos 2x \cos 3x} dx$ MII232
- (iii) $\int \frac{d\phi}{\sqrt{3} \cos \phi + \sin \phi}$ MII233
- (iv) $\int \frac{dx}{\cot \frac{x}{2} \cdot \cot \frac{x}{3} \cdot \cot \frac{x}{6}}$ MII234
6. (i) $\int \sin^2 x \cos^5 x dx$ MII235
- (ii) $\int \cos^3 x dx$ MII236
- (iii) $\int \sin^4 x dx$ MII237
- (iv) $\int \frac{dx}{\sqrt[4]{\sin^3 x \cos^5 x}}$ MII238
7. (i) $\int \frac{\sin^2 x}{\cos^4 x} dx$ MII239
- (ii) $\int \frac{\sqrt{\sin x} dx}{\cos^{5/2} x}$ MII240
- (iii) $\int \frac{dx}{\cos^4 x}$ MII241
- (iv) $\int \frac{dx}{\cos x \cdot \sin^3 x}$ MII242

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8. (i) $\int (\sin x)^{-11/3} (\cos x)^{-1/3} dx$ MII243
- (ii) $\int \sqrt[3]{\frac{\sin^2 x}{\cos^{14} x}} dx$ MII244
- (iii) $\int \frac{dx}{\cos^3 x \sqrt{\sin 2x}}$ MII245
9. (i) $\int \frac{dx}{1+9x^2}$ MII246
- (ii) $\int \frac{dx}{4x^2-9}$ MII247
- (iii) $\int \frac{dx}{2-3x^2}$ MII248
10. (i) $\int \frac{e^x dx}{e^{2x}+4}$ MII249
- (ii) $\int \frac{x^2}{9+16x^6} dx$ MII250
- (iii) $\int \frac{dx}{(x-1)^2+4}$ MII251
- (iv) $\int \frac{dx}{x^2+2x+3}$ MII252
11. (i) $\int \frac{dx}{4x^2+12x-7}$ MII253
- (ii) $\int \frac{dx}{x^2+x+1}$ MII254
- (iii) $\int \frac{\cos x dx}{3\sin^2 x+4\sin x+5}$ MII255
- (iv) $\int \frac{x dx}{x^4-3x^2+2}$ MII256

ANSWER KEY

1. (i) $\frac{1}{\cos x - 1} + c$ (ii) $3\sqrt[3]{\sin x} + C$ (iii) $2\sqrt{1 + \tan x} + C$ (iv) $\frac{(\tan x - x)^2}{2} + c$
2. (i) $-e^{\cot x} + e^{-\cot x} + C$ (ii) $\ln(1 + \sin x) + C$
 (iii) $\frac{1}{2}\tan^2 x + \ln|\cos x| + C$ (iv) $\frac{-1}{a+b} \frac{1}{(a\sin^2 x + b\cos^2 x)} + c$
3. (i) $\frac{\cos^2 x}{2} - \ln|\cos x| + C$ (ii) $\frac{1}{\sin x} - \frac{1}{3\sin^3 x} + C$
 (iii) $2\sqrt{\cos \alpha} \left(\frac{\cos^2 \alpha}{5} - 1 \right) + C$ (iv) $2\tan\left(\frac{x}{2} + \frac{\pi}{4}\right) - x + C$
4. (i) $\ln\left(\frac{\sin x}{x}\right) + C$ (ii) $\frac{1}{2}\ln|\sin(2x+1)| + C$ (iii) $C - \frac{1}{3}\ln|\cos 3x|$ (iv) $\ln\left|\frac{\sec x}{\cos \sec x + \cot x}\right| + c$
5. (i) $\ln\left|\frac{\sec x + \tan x}{\cos \sec x + \cot x}\right| + c$ (ii) $\frac{1}{3}\ln \sec 3x + \frac{1}{2}\ln \sec 2x + c$
 (iii) $\frac{1}{2}\ln\left|\tan\left(\frac{\phi}{2} + \frac{\pi}{6}\right)\right| + c$ (iv) $2\ln\left(\sec \frac{x}{2}\right) - 3\ln\left(\sec \frac{x}{3}\right) - 6\ln\left(\sec \frac{x}{6}\right) + C$
6. (i) $\frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C$ (ii) $\sin x - \frac{1}{3}\sin^3 x + C$
 (iii) $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$ (iv) $4\sqrt[4]{\tan x} + c$
7. (i) $\frac{1}{3}\tan^3 x + C$ (ii) $\frac{2}{3}\tan^{3/2} x + C$ (iii) $\tan x + \frac{\tan^3 x}{3} + c$ (iv) $\ln|\tan x| - \frac{1}{2\sin^2 x} + c$
8. (i) $\frac{-3}{8}\left(\frac{1+4\tan^2 x}{\tan^{\frac{2}{3}} x}\right) + c$ (ii) $\frac{3}{55}\tan^{\frac{5}{3}} x (11 + 5\tan^2 x) + c$ (iii) $\frac{\sqrt{2}}{5}\sqrt{\tan x}(5 + \tan^2 x) + c$
9. (i) $\frac{1}{3}\tan^{-1} 3x + c$ (ii) $\frac{1}{12}\ln\left|\frac{2x-3}{2x+3}\right| + C$ (iii) $\frac{1}{2\sqrt{6}}\ln\left|\frac{\sqrt{2}+x\sqrt{3}}{\sqrt{2}-x\sqrt{3}}\right| + C$
10. (i) $\frac{1}{2}\arctan \frac{e^x}{2} + C$ (ii) $\frac{1}{36}\tan^{-1}\left(\frac{4x^3}{3}\right) + C$ (iii) $\frac{1}{2}\arctan \frac{x-1}{2} + C$ (iv) $\frac{1}{\sqrt{2}}\arctan \frac{x+1}{\sqrt{2}} + C$
11. (i) $\frac{1}{16}\ln\left|\frac{2x-1}{2x+7}\right| + c$ (ii) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$
 (iii) $\frac{1}{\sqrt{11}}\tan^{-1} \frac{3t+2}{\sqrt{11}} + c$ (iv) $\ln\sqrt{\frac{x^2-2}{x^2-1}} + C$



Indefinite Integration : RACE-04

Subjective Type Question

Evaluate the following integrals :

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| 1. | (i) | $\int \frac{dx}{\sqrt{1-25x^2}}$ | MII257 |
| | (ii) | $\int \frac{dx}{\sqrt{4-9x^2}}$ | MII258 |
| 2. | (i) | $\int \frac{dx}{\sqrt{4x^2+9}}$ | MII259 |
| | (ii) | $\int \frac{x^{\frac{5}{2}}}{\sqrt{1+x^7}} dx$ | MII260 |
| 3. | (i) | $\int \frac{dx}{\sqrt{x^2+2x+10}}$ | MII261 |
| | (ii) | $\int \frac{dx}{\sqrt{9x^2+12x-5}}$ | MII262 |
| 4. | (i) | $\int \frac{dx}{\sqrt{1-(2x+3)^2}}$ | MII263 |
| | (ii) | $\int \frac{dx}{\sqrt{x-x^2}}$ | MII264 |
| 5. | (i) | $\int \frac{dx}{\sqrt{8+6x-9x^2}}$ | MII265 |
| | (ii) | $\int \frac{dx}{\sqrt{1+e^{-2x}}}$ | MII266 |
| 6. | (i) | $\int \frac{dx}{\sqrt{9+16\sec^2 x}}$ | MII267 |
| | (ii) | $\int \frac{2x+3}{\sqrt{1+x^2}} dx$ | MII268 |

7. (i) $\int \sqrt{\frac{1+x}{1-x}} dx$ MII269
 (ii) $\int \sqrt{\frac{x-3}{2-x}} dx$ MII270
8. (i) $\int \sqrt{6-x^2} dx$ MII271
 (ii) $\int \sqrt{9x^2-5} dx$ MII272
9. (i) $\int \sqrt{1-4x-x^2} dx$ MII273
 (ii) $\int \sqrt{1+3x-x^2} dx$ MII274
10. (i) $\int \sqrt{x^2+4x+6} dx$ MII275
 (ii) $\int \sqrt{1-\sin^4 x} dx$ MII276

ANSWER KEY

1. (i) $\frac{1}{5} \arcsin 5x + C$ (ii) $\frac{1}{3} \arcsin \frac{3x}{2} + C$
2. (i) $\frac{1}{2} \ln(2x + \sqrt{4x^2+9}) + c$ (ii) $\frac{2}{7} \ln \left| x^{\frac{7}{2}} + \sqrt{1+x^7} \right| + c$
3. (i) $\ln|x+1+\sqrt{x^2+2x+10}| + c$ (ii) $\frac{1}{3} \ln|3x+2+\sqrt{9x^2+12x-5}| + c$
4. (i) $\frac{1}{2} \arcsin(2x+3) + C$ (ii) $2 \arcsin \sqrt{x} + C$
5. (i) $\frac{1}{3} \arcsin \frac{3x-1}{3} + C$ (ii) $\ln|e^x + \sqrt{e^{2x}+1}| + c$
6. (i) $\frac{1}{3} \sin^{-1} \frac{3 \sin x}{5} + c$ (ii) $2\sqrt{1+x^2} + 3 \ln(x + \sqrt{1+x^2}) + c$
7. (i) $\arcsin x - \sqrt{1-x^2} + C$ (ii) $\sqrt{(x-2)(3-x)} - \sin^{-1} \sqrt{3-x} + C$
8. (i) $\frac{x}{2} \sqrt{6-x^2} + \frac{6}{2} \sin^{-1} \frac{x}{\sqrt{6}} + c$ (ii) $\frac{x}{2} \sqrt{9x^2-5} - \frac{5}{6} \ln \left| x + \sqrt{x^2 - \frac{5}{9}} \right| + c$
9. (i) $\frac{5}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{5}} \right) + \frac{x+2}{2} \sqrt{1-4x-x^2} + C$ (ii) $\frac{2x-3}{4} \sqrt{1+3x-x^2} + \frac{13}{8} \sin^{-1} \frac{2x-3}{\sqrt{13}} + c$
10. (i) $\frac{(x+2)}{2} \sqrt{x^2+4x+6} + \log|x+2+\sqrt{x^2+4x+6}| + C$
 (ii) $= \frac{\sin x}{2} \sqrt{1+\sin^2 x} + \frac{1}{2} \ln|\sin x + \sqrt{1+\sin^2 x}| + C$



Indefinite Integration : RACE-05

Subjective Type Question

Evaluate the following integrals :

- | | | | |
|----|------|---|--------|
| 1. | (i) | $\int \frac{(x+2)dx}{x^2+2x+2}$ | MII277 |
| | (ii) | $\int \frac{(3x-1)dx}{\sqrt{x^2+2x+2}}$ | MII278 |
| 2. | (i) | $\int \frac{5x+4}{\sqrt{x^2+4x+1}} dx$ | MII279 |
| | (ii) | $\int x\sqrt{x+x^2} dx$ | MII280 |
| 3. | (i) | $\int (x+1)\sqrt{2x^2+3} dx$ | MII281 |
| | (ii) | $\int \frac{x^2+2x+2}{x^2+1} dx$ | MII282 |
| 4. | (i) | $\int \frac{2x^2+5x+6}{\sqrt{x^2+1}} dx$ | MII283 |
| | (ii) | $\int \frac{x^2+1}{x^4-x^2+1} dx$ | MII284 |
| 5. | (i) | $\int \frac{1}{1+x^4} dx$ | MII285 |
| | (ii) | $\int \frac{(e^{3x}+e^x)dx}{e^{4x}-e^{2x}+1}$ | MII286 |
| 6. | (i) | $\int \frac{(2x^2-5)dx}{x^4-5x^2+6}$ | MII287 |
| | (ii) | $\int (\sqrt{\tan x} - \sqrt{\cot x}) dx$ | MII288 |

7. (i) $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx$ MII289
- (ii) $\int \frac{2+3x^2}{1+x^4} dx$ MII290
8. (i) $\int \sqrt{\tan x} dx$ MII291
- (ii) $\int \frac{x}{(x-3)\sqrt{x+1}} dx$ MII292
9. (i) $\int \frac{1}{(x^2-4)\sqrt{2x+1}} dx$ MII293
- (ii) $\int \frac{dx}{(1-x^2)\sqrt{2+x}}$ MII294
10. (i) $\int \frac{dx}{x\sqrt{x^2+4}}$ MII295
- (ii) $\int \frac{(x+\sqrt{1+x^2})^{15}}{\sqrt{1+x^2}}.dx$ MII296

ANSWER KEY

1. (i) $\frac{1}{2} \ln(x^2 + 2x + 2) + \arctan(x + 1) + c$ (ii) $3\sqrt{x^2 + 2x + 2} - 4\ln(x + 1 + \sqrt{x^2 + 2x + 2}) + c$
2. (i) $5\sqrt{x^2 + 4x + 1} - 6\ln|x + 2 + \sqrt{x^2 + 4x + 1}| + C$
 (ii) $\frac{(x + x^2)^{\frac{3}{2}}}{3} - \frac{1}{2} \left(\frac{x + \frac{1}{2}}{2} \sqrt{x^2 + x} - \frac{1}{4 \cdot 2} \ln \left(x + \frac{1}{2} + \sqrt{x^2 + x} \right) \right) + c$
3. (i) $\frac{1}{6} (2x^2 + 3)^{\frac{3}{2}} + \frac{x}{2} \sqrt{2x^2 + 3} + \frac{3\sqrt{2}}{4} \ln \left| x + \sqrt{x^2 + \frac{3}{2}} \right| + c$
 (ii) $x + \ln|x^2 + 1| + \tan^{-1} x + c$
4. (i) $x\sqrt{x^2 + 1} + 5\ln|x + \sqrt{x^2 + 1}| + 5\sqrt{x^2 + 1} + c$ (ii) $\tan^{-1} \left(\frac{x^2 - 1}{x} \right) + C$
5. (i) $\frac{1}{2\sqrt{2}} \left[\tan^{-1} \left(\frac{x^2 - 1}{\sqrt{2}x} \right) - \frac{1}{2} \ln \frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right]$ (ii) $\tan^{-1}(e^x - e^{-x}) + c$
6. (i) $\frac{1}{2\sqrt{2}} \ln \left| \frac{x - \sqrt{2}}{x + \sqrt{2}} \right| + \frac{1}{2\sqrt{3}} \ln \left| \frac{x - \sqrt{3}}{x + \sqrt{3}} \right| + c$ (ii) $\frac{1}{\sqrt{2}} \ln \left| \frac{\sqrt{\tan x} + \sqrt{\cot x} - \sqrt{2}}{\sqrt{\tan x} + \sqrt{\cot x} + \sqrt{2}} \right| + c$
7. (i) $2 \tan^{-1} \sqrt{x + \frac{1}{x} + 1} + C$ (ii) $\frac{5}{2\sqrt{2}} \tan^{-1} \frac{x - \frac{1}{x}}{\sqrt{2}} + \frac{1}{4\sqrt{2}} \ln \left| \frac{x + \frac{1}{x} - \sqrt{2}}{x + \frac{1}{x} + \sqrt{2}} \right| + C$
8. (i) $\frac{1}{\sqrt{2}} \tan^{-1} \frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} + \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{\tan x} + \sqrt{\cot x} - \sqrt{2}}{\sqrt{\tan x} + \sqrt{\cot x} + \sqrt{2}} \right| + C$
 (ii) $2\sqrt{x+1} + \frac{3}{2} \ln \left| \frac{\sqrt{x+1} - 2}{\sqrt{x+1} + 2} \right| + C$
9. (i) $\frac{1}{4\sqrt{5}} \ln \left| \frac{\sqrt{2x+1} - \sqrt{5}}{\sqrt{2x+1} + \sqrt{5}} \right| - \frac{1}{2\sqrt{3}} \tan^{-1} \frac{\sqrt{2x+1}}{\sqrt{3}} + C$
 (ii) $\frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{x+2} + \sqrt{3}}{\sqrt{x+2} - \sqrt{3}} \right) + \frac{1}{2} \ln \left(\frac{|\sqrt{x+2} - 1|}{\sqrt{x+2} + 1} \right) + c$
10. (i) $\frac{-1}{2} \ln \left| \frac{1}{x} + \sqrt{\frac{1}{x^2} + \frac{1}{4}} \right| + c$ (ii) $\frac{(x + \sqrt{1 + x^2})^{15}}{15} + c$



Indefinite Integration : RACE-06

Subjective Type Question

Evaluate the following integrals :

1. (i) $\int \frac{dx}{x^n (1+x^n)^{1/n}}$ MII297
 (ii) $\int \frac{dx}{x(x^2+1)}$ MII298
2. (i) $\int \frac{dx}{x^2(x^3+1)^{2/3}}$ MII299
 (ii) $\int \frac{dx}{x^3(x^3+1)^{1/3}}$ MII300
3. (i) $\int \frac{1+x^4}{(1-x^4)^{3/2}} dx$ MII301
 (ii) $\int \frac{2x-4}{x^2 \sqrt{x^2-2x+2}} dx$ MII302
4. (i) $\int \frac{dx}{x^2(x^5+1)^{4/5}}$ MII303
 (ii) $\int \frac{x \sin \alpha + 2}{(x^2 + x \sin \alpha + 1)^{3/2}} dx$ MII304
5. (i) $\int \frac{(x^4-x)^{1/4}}{x^5} dx$ MII305
 (ii) $\int \frac{x^2-1}{x^3 \sqrt{2x^4-2x^2+1}} dx$ MII306
6. (i) $\int \frac{dx}{x(x^4+1)(x^4+2)}$ MII307
 (ii) $\int x^{1/3} [1+x^{1/3}+x^{2/3}] [2x^{2/3}+3x^{1/3}+6]^3 dx$ MII308

Indefinite Integration

7. (i) $\int \frac{(2x^{12} + 5x^9)}{(x^5 + x^3 + 1)^3} dx$ MII309
- (ii) $\int (x^{7m} + x^{2m} + x^m)(2x^{6m} + 7x^m + 14)^{1/m} dx$ MII310
8. (i) $\int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$ MII311
- (ii) $\int \frac{\cos 2x \, dx}{1 + \sin x \cos x}$ MII312
9. (i) $\int \frac{\sin 2x \, dx}{4 - \cos^2 2x}$ MII313
- (ii) $\int \frac{dx}{1 + 4\sin^2 x}$ MII314
10. (i) $\int \frac{\sin x}{\sin x + \cos x} dx$ MII315
- (ii) $\int \frac{\sin x}{\sin 3x} dx$ MII316
11. (i) $\int \frac{dx}{4 - 3\cos^2 x + 5\sin^2 x}$ MII317
- (ii) $\int \frac{dx}{3\sin^2 x + \sin x \cos x + 1}$ MII318
12. (i) $\int \frac{dx}{3 + \sin x}$ MII319
- (ii) $\int \frac{dx}{5 - 4\sin x + 3\cos x}$ MII320
13. (i) $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx$ MII321
- (ii) $\int \frac{3\sin x + 2\cos x}{3\cos x + 2\sin x} dx$ MII322
14. (i) $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ MII323
- (ii) $\int \frac{dx}{\sin^4 x + \cos^4 x}$ MII324

ANSWER KEY

1. (i) $\frac{1}{(1-n)}\left(\frac{1}{x^n}+1\right)^{\frac{-1}{n}+1}+c$ (ii) $-\frac{1}{2}\ell n\left(1+\frac{1}{x^2}\right)+c$
2. (i) $-\left(1+\frac{1}{x^3}\right)^{\frac{1}{3}}+c$ (ii) $-\frac{1}{2}\left(1+\frac{1}{x^3}\right)^{\frac{2}{3}}+c$
3. (i) $\frac{1}{\left(\frac{1}{x^2}-x^2\right)^{1/2}}+C$ (ii) $\frac{2\sqrt{x^2-2x+2}}{x}+c$
4. (i) $-\left(1+\frac{1}{x^5}\right)^{\frac{1}{5}}+c$ (ii) $\frac{2}{\left(1+\frac{\sin \alpha}{x}+\frac{1}{x^2}\right)^{\frac{1}{2}}}+c$
5. (i) $\frac{4}{15}\left(1-\frac{1}{x^3}\right)^{\frac{5}{4}}+c$ (ii) $\frac{1}{2}\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$
6. (i) $-\frac{1}{8}\left(\ell n\frac{(x^4+1)^2}{x^8+2x^4}\right)+c$ (ii) $\frac{\left(2x+3x^{\frac{2}{3}}+6x^{\frac{1}{3}}\right)^4}{8}+c$
7. (i) $\frac{x^{10}}{2(x^5+x^3+1)^2}+c$ (ii) $\frac{1}{14(m+1)}(2x^{7m}+7x^{2m}+14x^m)^{\frac{m+1}{m}}+c$
8. (i) $\frac{1}{\left(\frac{1}{x^5}+\frac{1}{x^7}+2\right)}+c$ (ii) $\ell n(2+\sin 2x)+c$
9. (i) $-\frac{1}{8}\ell n\left|\frac{2+\cos 2x}{2-\cos 2x}\right|+c$ (ii) $\frac{1}{\sqrt{5}}\tan^{-1}(\sqrt{5}\tan x)+c$
10. (i) $\frac{1}{2}x-\frac{1}{2}\ell n|\sin x+\cos x|+c$ (ii) $-\frac{1}{2\sqrt{3}}\ell n\left|\frac{\tan x-\sqrt{3}}{\tan x+\sqrt{3}}\right|+c$
11. (i) $\frac{1}{3}\tan^{-1}(3\tan x)+c$ (ii) $\frac{2}{\sqrt{15}}\tan^{-1}\left(\frac{8\tan x+1}{\sqrt{15}}\right)+c$
12. (i) $\frac{1}{\sqrt{2}}\tan^{-1}\frac{3\tan\frac{x}{2}+1}{2\sqrt{2}}+c$ (ii) $\frac{-1}{\tan\frac{x}{2}-2}+C$
13. (i) $\sqrt{2}\sin^{-1}(\sin x-\cos x)+c$ (ii) $\frac{7}{13}\times\frac{-5}{13}\ell n|3\cos x+2\sin x|+c$
14. (i) $-\frac{1}{(\sec x+\tan x)^{\frac{11}{2}}\left(\frac{1}{11}+\frac{(\sec x+\tan x)^2}{7}\right)}+c$ (ii) $\frac{1}{\sqrt{2}}\tan^{-1}\frac{(\tan^{-1}x-\cot x)}{\sqrt{2}}+C$



Indefinite Integration : RACE-07

Subjective Type Question

Evaluate the following integrals :

- | | | | |
|----|------|--|--------|
| 1. | (i) | $\int \frac{3x+2}{(x+1)(x+3)} dx$ | MII325 |
| | (ii) | $\int \frac{1}{(x+1)(x+2)(x+3)} dx$ | MII326 |
| 2. | (i) | $\int \frac{dx}{6x^3 - 7x^2 - 3x}$ | MII327 |
| | (ii) | $\int \frac{xdx}{(x+1)^3}$ | MII328 |
| 3. | (i) | $\int \frac{4x+3}{(x-2)^3} dx$ | MII329 |
| | (ii) | $\int \frac{x^2 dx}{(x+2)^2 (x+4)^2}$ | MII330 |
| 4. | (i) | $\int \frac{x^2 - 1}{(x+1)(x+2)^2} dx$ | MII331 |
| | (ii) | $\int \frac{1}{(x^2+1)(x+3)} dx$ | MII332 |
| 5. | (i) | $\int \frac{x^2 dx}{1-x^4}$ | MII333 |
| | (ii) | $\int \frac{3x+2}{(x+1)^2 (x+2)} dx$ | MII334 |
| 6. | (i) | $\int \frac{xdx}{x^3-1}$ | MII335 |
| | (ii) | $\int \frac{x^4 dx}{x^{15}-1}$ | MII336 |

- | | | | |
|-----|------|--|--------|
| 7. | (i) | $\int x \ln x \, dx$ | MII337 |
| | (ii) | $\int x e^{-x} \, dx$ | MII338 |
| 8. | (i) | $\int 2x^3 e^{x^2} \, dx$ | MII339 |
| | (ii) | $\int x^3 \sin(x^2) \, dx$ | MII340 |
| 9. | (i) | $\int \frac{\ln^3 x}{x^2} \, dx$ | MII341 |
| | (ii) | $\int \frac{\arcsin \sqrt{x}}{\sqrt{1-x}} \, dx$ | MII342 |
| 10. | (i) | $\int \tan^{-1} x \, dx$ | MII343 |
| | (ii) | $\int \frac{x^2 \tan^{-1} x}{1+x^2} \, dx$ | MII344 |
| 11. | (i) | $\int \arctan \sqrt{x} \, dx$ | MII345 |
| | (ii) | $\int \ln(x + \sqrt{1+x^2}) \, dx$ | MII346 |
| 12. | (i) | $\int \sec^3 x \, dx$ | MII347 |
| | (ii) | $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} \, dx$ | MII348 |
| 13. | (i) | $\int x^3 \sin x \, dx$ | MII349 |
| | (ii) | $\int (x^2 - 2x + 3) \cos 3x \, dx$ | MII350 |

ANSWER KEY

1. (i) $\frac{-1}{2} \ln|x+1| + \frac{7}{2} \ln|x+3| + C$ (ii) $\frac{1}{2} \ln|x+1| - \ln|x+2| + \frac{1}{2} \ln|x+3| + C$
2. (i) $\frac{-1}{3} \ln x + \frac{2}{33} \ln \left| x - \frac{3}{2} \right| + \frac{3}{11} \ln \left| x + \frac{1}{3} \right| + C$ (ii) $\frac{-1}{x+1} + \frac{1}{2(x+1)^2} dx$
3. (i) $\frac{-4}{x-2} - \frac{11}{2(x-2)^2} + C$ (ii) $2 \ln \left| \frac{x+4}{x+2} \right| - \frac{5x+12}{x^2+6x+8} + C$
4. (i) $\ln|x+2| + \frac{3}{x+2} + C$ (ii) $\frac{-1}{20} \ln|x^2+1| + \frac{3}{10} \tan^{-1} x + \frac{1}{10} \ln|x+3| + C$
5. (i) $-\frac{1}{2} \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + \frac{1}{2} \tan^{-1} x + C$ (ii) $4 \ln|x+1| + \frac{1}{x+1} - 4 \ln|x+2| + C$
6. (i) $= \frac{1}{3} \ln|x-1| - \frac{1}{6} \ln|x^2+x+1| - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$
 (ii) $\frac{1}{15} \left[\frac{1}{2} \ln \frac{(t-1)^2}{t^2+t+1} - \sqrt{3} \tan^{-1} \frac{2t+1}{\sqrt{3}} \right] + C$
7. (i) $\frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{x^2}{2} + C$ (ii) $-x.e^{-x} - e^{-x} + C$
8. (i) $(x^2-1)e^{x^2} + C$ (ii) $\frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + C$
9. (i) $-\frac{1}{x} \left[(\ln x)^2 + 2 \ln x + 2 \right] + C$ (ii) $-2 \sin^{-1} \sqrt{x} \cdot (\sqrt{1-x}) + 2\sqrt{x} + C$
10. (i) $x \cdot \tan^{-1} x + \ln \sqrt{1+x^2} + C$ (ii) $x \tan^{-1} x - (\tan^{-1} x)^2 - \ln \sqrt{1+x^2} + \frac{(\tan^{-1} x)^2}{2} + C$
11. (i) $x \tan^{-1} \sqrt{x} - \sqrt{x} + \tan^{-1} \sqrt{x} + C$ (ii) $x \ln(x + \sqrt{1+x^2}) - \sqrt{1+x^2} + C$
12. (i) $\frac{\sec x \tan x}{2} + \frac{1}{2} \ln |\sec x + \tan x| + C$ (ii) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \ln \sqrt{1-x^2} + C$
13. (i) $3(x^2-2) \sin x - (x^3-6x) \cos x + C$ (ii) $\frac{1}{27} [(9x^2-18x+25) \sin 3x + 3(2x-2) \cos 3x] + C$



Indefinite Integration : RACE-08

Subjective Type Question

Evaluate the following integrals :

1. (i) $\int \frac{e^x}{x} (1 + x \cdot \ln x) dx$ MII351
 (ii) $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) dx$ MII352
2. (i) $\int \frac{e^x (x-1)}{(x+1)^3} dx$ MII353
 (ii) $\int x e^{x^2} (\sin x^2 + \cos x^2) dx$ MII354
3. (i) $\int \frac{e^{2x} (\sin 4x - 2)}{1 - \cos 4x} dx$ MII355
 (ii) $\int e^x [\ln(\sec x + \tan x) + \sec x] dx$ MII356
4. (i) $\int \frac{e^{\sin^{-1} x}}{2x\sqrt{x-x^3}} (2x - \sqrt{1-x^2}) dx$ MII357
 (ii) $\int (\ell n x + 1) dx$ MII358
5. (i) $\int \sin(\ell n x) dx$ MII359
 (ii) $\int \left[\ln(\ln x) + \frac{1}{(\ln x)^2} \right] dx$ MII360
6. (i) $\int \frac{\ln x}{(\ln x + 1)^2} dx$ MII361
 (ii) $\int e^{x/2} \sin\left(\frac{x}{2} + \frac{\pi}{4}\right) dx$ MII362

Indefinite Integration

7. (i) $\int e^{-2x} \sin 3x dx$ MII363
- (ii) $\int e^{\sin x} \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$ MII364
8. (i) $\int \frac{(x + \sin x)^2 + (1 + \cos x)^2}{(x + \sin x)(1 + \cos x)} dx$ MII365
- (ii) $\int \frac{2x - \sin 2x}{1 - \cos 2x} dx$ MII366
- (iii) $\int \left(3x^2 \tan\left(\frac{1}{x}\right) - x \sec^2\left(\frac{1}{x}\right) \right) dx$ MII367
9. (i) If $I_n = \int \frac{x^n dx}{\sqrt{x^2 + a}}$, then prove that $I_n + \frac{n-1}{n} a I_{n-2} = \frac{1}{n} x^{n-1} \sqrt{x^2 + a}$ MII368
- (ii) If $I_n = \int x \operatorname{cosec}^n x dx$, $n \geq 3$, then prove that
- $$I_n = -\frac{1}{n-1} \cdot \operatorname{cosec}^{n-2} x \cdot \left(x \cot x + \frac{1}{n-2} \right) + \frac{n-2}{n-1} I_{n-2} + \frac{C}{n-1}$$
- MII369
- (iii) Prove that $\int \sin^p x \cos^q x dx = \frac{\sin^{p+1} x \cos^{q+1} x}{p+q} + \frac{q-1}{p+q} \int \sin^p x \cos^{q-2} x dx$ MII370
10. (i) $\int \frac{dx}{\sqrt[4]{(x-1)^3 (x+2)^5}}$ MII371
- (ii) $\int \frac{dx}{(x+1)^{6/5} (x-3)^{4/5}}$ MII372

ANSWER KEY

1. (i) $e^x \cdot \ln x + C$ (ii) $e^x \tan^{-1} x + C$
2. (i) $\frac{e^x}{(x+1)^2} + C$ (ii) $\frac{1}{2} e^{x^2} \sin(x^2) + C$
3. (i) $\frac{1}{2} e^{2x} \cot 2x + C$ (ii) $e^x \ln(\sec x + \tan x) + C$
4. (i) $\frac{e^{\sin^{-1} x}}{\sqrt{x}} + C$ (ii) $x \ln x + C$
5. (i) $\frac{x}{2} (\sin \ln x - \cos \ln x) + C$ (ii) $x \left(\ln(\ln x) - \frac{1}{\ln x} \right) + C$
6. (i) $\frac{x}{\ln x + 1} + C$ (ii) $\frac{e^{x/2} \sin \frac{x}{2}}{\sqrt{2}} + C$
7. (i) $I = -\frac{e^{-2x}}{13} (3 \cos 3x + 2 \sin 3x) + C$ (ii) $e^{\sin x} (x - \sec x) + C$
8. (i) $x \tan \frac{x}{2} + \ln |x + \sin x| + C$ (ii) $-x \cot x + C$ (iii) $x^3 \tan \frac{1}{x} + C$
9. (i) $I_n = \frac{x^{n-1} \sqrt{x^2 + a}}{n} - \frac{(n-1)}{n} a I_{n-2}$ (ii) $I_n = \left(\frac{n-2}{n-1} \right) I_{n-2} - \frac{\operatorname{cosec}^{n-2} x}{n-1} \left(x \cot x + \frac{1}{n-2} \right) + \frac{c(n-2)}{n-1}$
 (iii) $I = \frac{(\cos x)^{q-1} (\sin x)^{p+1}}{p+q} + \frac{q-1}{p+q} \int (\cos x)^{q-2} (\sin x)^p dx$
10. (i) $\frac{4}{3} \sqrt[4]{\frac{x-1}{x+2}} + C$ (ii) $\frac{5}{4} \left(\frac{x-3}{x+1} \right)^{1/5} + C$